

MAT 3150 — Homework 3
Problem 1 Walkthrough: Reverse Triangle Inequality

Goal. Prove that for any $x, y \in \mathbb{R}$,

$$\left| |x| - |y| \right| \leq |x - y|.$$

Tools.

- Triangle inequality: $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
- For $c \geq 0$: $|a| \leq c \iff -c \leq a \leq c$.

Step 1 (worked)

Start from the identity

$$x = (x - y) + y.$$

Take absolute values and apply the triangle inequality:

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

Subtract $|y|$ from both sides:

$$|x| - |y| \leq |x - y|. \tag{*}$$

Step 2 (your turn)

Do the same thing, but start from

$$y = (y - x) + x.$$

- What inequality do you get for $|y| - |x|$?
- Remember that $|y - x| = |x - y|$.

Step 3 (combine)

Let $a = |x| - |y|$ and $c = |x - y|$. Note that $c \geq 0$.

- Step 1 shows $a \leq c$.
- Step 2 should show $-a \leq c$, i.e. $a \geq -c$.

So $-c \leq a \leq c$. Which of the tools above turns that into a statement about $|a|$? Write that conclusion in terms of x and y .

Checklist for the final write-up

- Both one-sided inequalities are stated and justified by the triangle inequality.
- You explicitly use $|y - x| = |x - y|$.
- You cite the fact $|a| \leq c \iff -c \leq a \leq c$ (with $c \geq 0$) to finish.